



Senolytics in Brain Disorders: A Novel Pharmacological Approach to Control Cellular Senescence and Oxidative Stress

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Abstract

Cellular senescence is considered to play an important role in age-related disorders in humans. Various studies indicate that brain cells also undergo cellular senescence via multiple mechanisms such as metabolic dysfunction, inflammation, oxidative stress, and alterations of proteins, thereby contributing to neuronal loss in age-related neurodegenerative and other disorders. Hence, suppressing the burden of cell senescence via pharmacological agents known as senolytics could offer great promise for the management of such conditions. There are reports that various senolytics acting via various mechanisms have been studied in animal models of CNS disorders and in human postmortem brain tissues. These agents such as dasatinib, quercetin, and fisetin have shown some positive results in decreasing the load of cell senescence in a few CNS disorders. In this review, the impact of cell senescence in various brain disorders and the studies done on senolytics so far on CNS disorders have been reviewed.

Keywords

Cell senescence · Senolytics · Dasatinib · Quercetin · Fisetin

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20.1 Introduction

In 1961, human subcultured cells showed replicative and irreversible arrest of the cell cycle (Hickson et al. 2019). This phenomenon was termed senescence which is derived from the Latin word, *Senex*, which means “growing old.” Cellular senescence has been shown to have both beneficial and deleterious effects on organisms (Kang et al. 2015). The cellular response may differ depending on the intensity and type of stress and cell type, which may range from repair to cell death (Sapieha and Mallette 2018). It is not only a response to stress, but several stimuli such as radiation, chemotherapeutic agents, and oxidative stress impose cellular senescence in cells. Telomere damage, metabolic dysfunction, alterations of proteins, inflammation, stress, oncogenes, nucleotide damage, microbial damage, and mitogens also drive a cell into senescence (Mohamad Kamal et al. 2020). Similar to cellular fate, senescence also involves differentiation, proliferation, necrosis, and apoptosis of cells. Senescent cells release growth factors and cytokines and are metabolically active cells that play a critical role in tissue homeostasis, embryogenesis, tissue patterning, and wound healing. Senescence modifies gene expression, alters the function of cell organelles, modifies histones, and hence leads to significant metabolic and morphological alterations (Calcinotto et al. 2019; Ogrodnik et al. 2019; Ogrodnik 2021). Chronic stress can cause senescent cells to accumulate. When they accumulate, they are capable to cause destruction of tissue, dysfunction of progenitor and stem cells, and inhibition of immune system along with inflammation (both local and systemic). Elevation in number of senescent cells contributes toward several pathological conditions such as type 2 diabetes mellitus, liver diseases, atherosclerosis, osteoarthritis, and idiopathic pulmonary fibrosis. Thus, senescence is favorable for acute stress-induced cellular damage but has deleterious effects on chronic stress-induced disorders (Ogrodnik et al. 2019; Wissler Gerdes et al. 2020). Besides arrest of cell cycle, senescent cells also undergo functional, morphological, and biochemical changes (Sedelnikova et al. 2004; Bernadotte et al. 2016). The cells appear larger in size, cell survival signaling pathways get activated, and formation of heterochromatin foci occurs along with increase in expression of growth factors, pro-inflammatory cytokines, and metalloproteinases (Si et al. 2021).

Human body's natural defense mechanism identifies senescent cells through helper T-cells and is then removed by natural killer cells, lymphocytes, and macrophages (Han et al. 2003; Akbar and Henson 2011; Kowald et al. 2020). Multiple transcription factors like p53/CDKN1A (cyclin-dependent kinase inhibitor) (p21), CDKN2A (p16)/RB (retinoblastoma) (Shay et al. 1991; Hernandez-Segura et al. 2018; Casella et al. 2019), and p16^{INK4a} (Robles and Adami 1998; Calcinotto et al. 2019), p21^{CIP1}, p27, p19^{ARF}, p53, and PAI-1 are tumor suppressors that have considerable role in cellular senescence (Calcinotto and Alimonti 2017). p53 regulates the expression of p21 transcriptionally (El-Deiry et al. 1993). There is upregulation of p21 and p16 and increased expressions of CDKN in senescent cells (Stein et al. 1999). Further, elevated activity of senescence-associated β -galactosidase (SA- β gal⁺) at pH 6 is observed during cellular senescence (Hernandez-Segura et al. 2018; Casella et al. 2019). GATA4 (a protein-coding

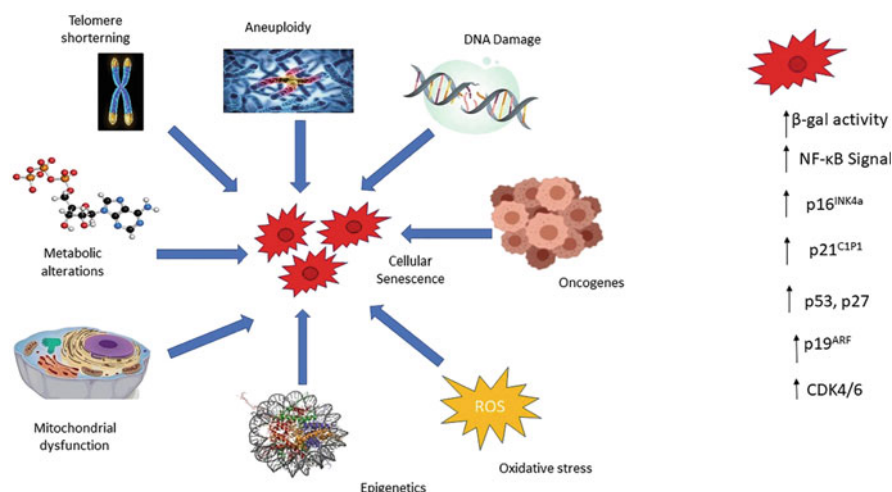


Fig. 20.1 Factors and mechanisms leading to cellular senescence. Various factors such as telomere shortening, aneuploidy, damage of DNA, development of oncogenes, development of reactive oxygen species (ROS) through oxidative stress, dysfunctional epigenetics, mitochondrial dysfunction, and metabolic alterations can drive cell into senescence. There is activation of beta-galactosidase activity that activates p53/p21^{CIP1} and p16^{INK4a}/RB networks. There is upregulation of NF-κB (nuclear factor kappa B) and cyclin-dependent kinase (CDK) functions

gene), a zinc finger transcription factor, is another important regulator of senescence. It acts as a link between DNA damage and autophagy that leads to inflammation and senescence by activating NF-κB through IL1A and TRAF3IP2 and serves several roles in multiple organs. It functions independently of p53 and p16^{INK4a} (Kang et al. 2015). Nuclear factor kappa B (NF-κB) pathway has been reported to promote senescence-associated secretory phenotype (SASP)-related cellular senescence. Shao et al. also revealed that Nrf-2 is another crucial antioxidant transcription factor (Shao et al. 2021). Markers of SASP-induced senescence include telomere-linked DNA damage foci, elevated response of lysosomal SA-β gal⁺, senescence-linked heterochromatin foci, and increased level of p21^{CIP1} and p16^{INK4a} (Ben-Porath and Weinberg 2005; Krishnamurthy et al. 2006). Figure 20.1 describes the various factors responsible for cellular senescence.

According to Costa and team, deficiency of paraoxonase 2 (PON2) leads to mitochondrial dysfunction. PON2 is found in several dopaminergic regions such as substantia nigra and substantia striatum; brain stem is an antioxidant and anti-inflammatory protein that binds to coenzyme Q10 in mitochondrial complex (Costa et al. 2014). Accumulation of senescent cells accelerates with process of normal aging in animals and humans. Permanent cell cycle arrest by senescent cells is regulated by p53-p21-RB and p16^{INK4A}. Initially, cell cycle is arrested due to higher levels of p53 that upregulates CDKN(p21), and arrest is mediated by cyclin-dependent kinase 4 (CDK4) and cyclin-dependent kinase 6 (CDK6) which leads to hypo-phosphorylation of RB protein and arrests S phase of cell cycle. Apart from

NF- κ B-SASP, other markers of cellular senescence include cyclic GMP-AMP synthase (cGAS), p38 mitogen-activated protein kinases (p38MAPK), stimulator of interferon genes (STING), mammalian target of rapamycin (mTOR), and NOTCH signaling pathways (Kritsilis et al. 2018; Watanabe et al. 2017; González-Gualda et al. 2021).

SASP is developed by most cells and consists of chemokines, growth factors, procoagulant factors, pro-inflammatory chemokines such as interleukins (IL-1 β and IL-6), proteases bioactive lipids, miRNAs, micro-vesicles, and nucleotides (Xu et al. 2018; Martel et al. 2020). In addition to this, some chemotherapeutic agents such as doxorubicin and etoposide, exposure to ionizing radiation, and oxidative stress also contribute to senescence (Zhang et al. 2021).

There are SCAPs (senescent cell anti-apoptotic pathways) that inhibit the death of senescent cells in human pre-adipocytes. SCAPs pathways include ephrins/dependence receptors, PI3K δ /Akt, Bcl-2/B-cell, p53/FOXO4a/p21/serpine, HIF-1 α , and HSP-90 pathway (Wissler Gerdes et al. 2020).

Progressive research to identify drugs that can selectively eliminate senescent cells without affecting non-senescent cells leads to the development of senolytics. These are the class of drugs that help in removal or suppression of senescence of cells pharmacologically by inducing apoptosis of senescent cells. Senolytics do not exhibit their intrinsic activity through any biochemical pathway or receptors or enzymes rather they directly affect senescent cells by critically targeting them. Approach toward the process of senolysis seems promising for good health, age-related disorders, and prolonged life (Wissler Gerdes et al. 2020; Zhang et al. 2021). Senolytics promote apoptosis of senescent cells through SASP. Quercetin (Q) and dasatinib (D) are potent senolytics that suppress the senescent cell anti-apoptotic pathway network. Q is more selective as it targets Bcl-2 family, HIF-1 α , PI3K, and p21-related anti-apoptotic pathways (Hickson et al. 2019). Figure 20.2 presents the overview of mechanism of senolytics action.

There are several other drugs/compounds that exhibit senolytic or senotherapeutic activities in several disorders. The first-generation senolytics described in 2015 include dasatinib (kinase inhibitor) and quercetin. After that, many classes of senolytics have been described like natural compounds (fisetin, piperlongumine, curcumin, ouabain), Bcl-2 family inhibitors/Bcl-2 homology 3 (BH3) mimetics (navitoclax/ABT263, ABT-737, A1331852 and A1155463, UBX1325), inhibitors of MDM2/p53 interaction (UBX0101, P5091), Hsp90 inhibitors (geldanamycin, tanespmycin, alvespmycin/17-DMAG), p53-binding inhibitors (FOXO4-DRI), and histone deacetylase (HDAC) inhibitors (Panobinostat) (Kim and Kim 2019; Lagoumtzi and Chondrogianni 2021).

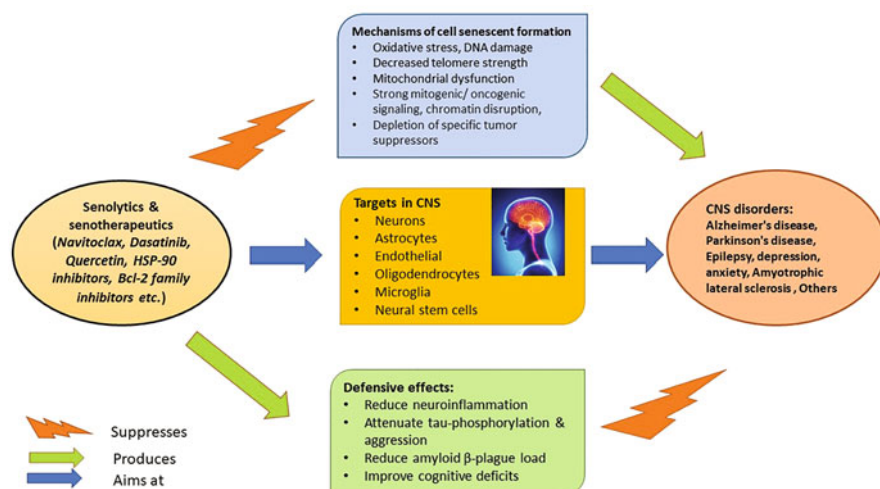


Fig. 20.2 Overview of senolytics on mechanisms inducing cell senescence and their effects on CNS disorders

20.2 The Role of Cell Senescence and Senolytics in CNS Disorders

Cellular senescence is associated with several CNS disorders. Preclinical and clinical research studies to substantiate their role in therapy of diseases have been conducted to substantiate their role.

Table 20.1 enlists various clinical trials regarding the role of senolytics in CNS disorders.

20.2.1 Anxiety

Psychological stress has a direct relation with increased oxidative stress, lower telomerase activity, and reduced telomere length; hence, the patients with depression, anxiety, or stress disorders have shorter telomere length than healthy controls (Wang et al. 2017). Few studies have reported that antianxiety and antidepressant drug therapy shows neuroprotective effect which is related to the reversal of aging-induced hallmarks like telomere length (Zhou et al. 2011; Perna et al. 2016).

Nociceptive opioid receptor agonist MCOPPB (1-[1-(1-methylcyclooctyl)-4-piperidinyl]-2-[(3R)-3-piperidinyl]-1H-benzimidazole), a potent anxiolytic, recently demonstrated its senolytic potential. It increased lipid storage in male C57/BL6 mice and *E. elegans* worm indicating longevity. Additionally, it showed tissue-specific cellular senescence in peripheral tissues of mice. Further, it significantly decreased

Table 20.1 Clinical trials on senolytics in CNS disorders

S. No.	Official title	Clinical trial registration number	Drugs and phases
1	An open-label pilot study of Senolytics for Alzheimer's disease	NCT04785300	Dasatinib + quercetin Phase 1, phase 2
2	Pilot study to investigate the safety and feasibility of senolytic therapy to modulate progression of Alzheimer's disease (SToMP-AD)	NCT04063124	Dasatinib + quercetin Phase 1, phase 2
3	Phase II clinical trial to evaluate the safety and feasibility of senolytic therapy in Alzheimer's disease	NCT04685590	Dasatinib + quercetin Phase 2
4	A multi-domain lifestyle intervention protocol to detect changes in retinal amyloid among individuals with mild cognitive impairment	NCT02741804	Fisetin + others Phase NA
5	Impact of the combined treatment of curcumin and resveratrol liposomed polyphenols with G04CB02 on the clinical improvement of ALS patients	NCT04654689	Curcumin Phase 2
6	A phase 2, prospective, multicenter, randomized, double-masked, active-controlled study to assess the safety, tolerability, and evidence of activity of a repeat intravitreal injection of UBX1325 in patients with neovascular AMD	NCT05275205	UBX1325 Phase 2

the number of senescent cells in DOX-treated Huh-7 and HepG2 cells (Raffaele et al. 2022).

Metabolic disorder like obesity is also related to augmented senescent cell burden and neuropsychiatric disorders like anxiety and depression (Garipey et al. 2010; Hryhorczuk et al. 2013; Palmer et al. 2022). To evaluate these effects, Ogrodnik and team proved that dasatinib and quercetin treatment improved high-fat diet (HFD)-induced obesity, anxiety, and senescence in 8-month-old C57Bl/6 mice. As accumulation of lipids is crucial for senescence, they investigated the role of glial senescent cells in obesity-associated neurogenesis and anxiety-like behavior in vivo. Elevated plus maze model was used to evaluate the anxiety-like behavior in mice. Perigonadal adipose tissue was measured as a senescent marker. It was revealed that removal of senescent cells reduced the obesity-associated anxiety-like behavior in the animal model. HFD-treated mice displayed proximity of senescent Plin2⁺ glial cells that are also expressed marker of neurons and neuronal precursor cells, doublecortin (Dcx). Senolytic therapy also upregulated Dcx, CD133⁺, and Nestin⁺ that are markers of immature neurons, ependymal cells, and neural precursor cells, respectively (Ogrodnik et al. 2019).

20.2.2 Depression

Clearing of senescent cells could provide new avenues for stress and depressive illness. To prove this theory, Lin and co-workers studied SA- β -gal activity of active senescent cells in vivo in C57BL/6 male mice. Behavioral despair and depression in animals were studied with forced swim test and chronic unpredictable stress (CUS) model. CUS was employed for 1 month with different stressors such as cage tilting, sleep deprivation, food/water deprivation, rapid light and dark cycles, restraints, removal of bedding, and water at the bottom of the cages. Results showed that CUS-treated mice exhibited higher levels of p16^{INK4a} levels indicating cellular senescence. The outcome also revealed that dasatinib and quercetin treatment decreased the load of SA- β -gal-type cells and expression of p16^{INK4a} in mice hippocampal region (Lin et al. 2021).

Effect of fisetin was evaluated in X-ray (10 Gy)-induced depression in female MRL/lpr systemic lupus erythematosus (SLE) and haplotype female MRI/MPJ mice. Oral administration of fisetin decreased the proximity of senescent cells in CA3 region, part of hippocampus associated with depression and presence of SA- β -gal⁺ cells. Further, transcription of SASP genes at the hippocampal region in SLE mouse and attenuation of depression-like behavior in MRL/lpr mouse model was seen as well (Saito et al. 2021).

20.2.3 Alzheimer's Disease

Neurodegenerative disorders like Alzheimer's disease (AD) are outcome of imbalanced autophagy and lysosomal dysfunction. Cellular senescence also contributes toward pathology of AD and ascertain age, susceptibility, and rate of progression of disease (Saez-Atienzar and Masliah 2020). Major mechanisms of cellular senescence involved in neurodegeneration includes chronic inflammation, dysfunctional regenerative capacities of brain cerebral hypoperfusion leading to functional loss altered blood brain barrier and erratic re-entry of differentiated post-mitotic neurons. Higher expression of key regulators of cell cycle dysfunction and cyclin-dependent kinase inhibitor p21^{CIP1} and p53 along with raised activity of p38MAPK activity is also found. Additionally, elevated levels of TNF- α , interleukins (IL-6 and IL-1 β), transforming growth factor- β (TGF- β), and γ H2AX were found in hippocampus of AD patients (Kritsilis et al. 2018).

Curraise et al. investigated the role of oral fisetin (25 mg/kg/day) on learning and memory in double transgenic APPswe/PS1dE9 AD mice (3–12 months old). The results showed that fisetin elevated extracellular signal-regulated kinase (ERK) phosphorylation and reduced protein carbonylation suggesting its role in oxidative stress. Further, it also decreased the expression of p25 (cyclin-dependent kinase 5 activator) in brains of mice, which is responsible for neurodegeneration and neuroinflammation (Currais et al. 2014).

Fisetin treatment remarkably suppressed phosphorylation of tau protein, levels of BACE-1, and A β accumulation in A β _{1–42} (3 μ L/5 min/mouse)-treated mice. Fisetin

also reversed the A β_{1-42} -induced elevated synaptic proteins such as SYN, SNAP-25, SNAP-23, post-synaptic density protein 95 (PSD-95), phospho-CREB (p-CREB), phospho-GLuR1 (p-GluR1), and phospho-calcium/calmodulin-stimulated protein kinase II (p-CAMKII). Additionally, it also increased levels of p-Akt, p-PI3K, and p-GSK3 β in A β_{1-42} mice and suppressed inflammation in neurons and apoptotic neurodegeneration (Ahmad et al. 2017).

Zhang et al. evaluated the impact of Q (50 m/kg) and D (12 mg/kg) for 9 days on APP/PS1 mouse model after oral administration. Since, A β plaque is responsible for neuritic degeneration, local inflammation, and lysosomal dysfunction. Hence, they also quantified A β plaque and associated oligodendrocyte progenitor cells (OPC) as they were accumulated in the regions of brain in AD patients. The results showed that senescent cells in AD patients and A β mouse model of AD-expressed OPCs transform into pro-inflammatory phenotypes of senescent cells and accumulation of A β plaque also induces cellular senescence. Senolytic treatment cleared the p16 cellular expression of OPC cells in the mouse model and decreased the microglial activation and expression of pro-inflammatory cytokines as well. Senolytic treatment also improved the patterns of learning and enhanced memory in APP/PS1 AD mouse model affirming its substantial role in cellular senescence responsible for AD (Zhang et al. 2019).

Fatt et al. reported that administration of navitoclax (ABT-263), *in vivo* in middle-aged mice, caused ablation of senescent cells at hippocampus after a month therapy. This in turn rapidly increased the expression of neural precursor cells at the hippocampus. Further, after a year of therapy, there was an increase in SA- β -Gal-positive in the sub-granular zone by 3 times. Additionally, navitoclax also exhibited an improvement in spatial learning and memory when tested through the Morris water maze (Fatt et al. 2022). Navitoclax also reversed the cell senescence in irradiation (5 Gy)-induced senescence in brain astrocytes of transgenic p16-3MR mice (Yabluchanskiy et al. 2020).

Studies suggested that progressive telomere shortening is related to biological aging and age-related morbidities, weakness, and mortality (Whittemore et al. 2019; Zhu et al. 2019). Further, it is also shown to activate cellular senescence along with inflammation and mitochondria (Zhu et al. 2019). Metformin via multiple mechanisms is shown to reduce telomere shortening in diabetic patients and also encourage protective feedback mechanisms on the maintenance of telomere in addition to the improvement in glucose tolerance and insulin sensitivity (Kulkarni et al. 2020).

20.2.4 Parkinson's Disease (PD)

The damage of neurons from the substantia nigra and the build-up of ubiquitinated alpha synuclein along with Lewy bodies and Lewy neuritis in cytoplasmic inclusion and within neurites, respectively, are the characteristic features of PD (Kritsilis et al. 2018). Many studies documented involvement of cellular senescence in the pathogenesis of neurodegenerative disorders including PD (Lee et al. 2021). The

expression of p16^{INK4a} mRNA was increased in brain samples of PD patients. Senescent cell markers are also present within astrocytes of brain tissues in postmortem PD samples. These samples exhibited elevated levels of pro-inflammatory SASP factors, including IL-6, IL-8, and IL- α (Chinta et al. 2018).

Paraquat, a toxic compound capable of inducing PD, can also stimulate cellular senescence and generate the production of SASP factors in astrocytes. This action leads to neuronal cell death. When these senescent astrocytes were removed in *in vitro* study, it reversed the paraquat-induced PD (Chinta et al. 2018).

In the pathogenesis of PD, mitochondrial hyperactivity has been reported as an early event. Approaches that decrease mitochondrial respiration may serve as new strategy for the management of PD. In this regard, antidiabetic drug metformin seems to be a promising agent. It improves motor functions and neuronal viability by decreasing mitochondrial respiration (Mor et al. 2020). Hence, reducing mitochondrial respiration could be employed as a strategy to explore new treatment approach in PD.

Quercetin has ability to bind with α -synuclein through a covalent bond (Singh et al. 2020). The neuroprotective and anti-PD roles of quercetin were observed in primary cultured rat cortical cells against oxidative neuronal injury and on 6-hydroxydopamine-induced rat model, respectively (Haleagrahara et al. 2011; Bakoyiannis et al. 2019). The mechanisms suggested for neuroprotective effect of quercetin involves activation of sirtuins (SIRT1) and induction of autophagy (Kida and Goligorsky 2016). Quercetin alters the cellular senescence via Nrf2 pathway as it suppresses SASP-induced cellular senescence (Stein et al. 1999). Along with induction of Nrf2-ARE (antioxidant response element), it also induces PON2 (Acikara et al. 2022).

20.2.5 Epilepsy

Among the various biomarkers indicated for cell senescence, neurodegeneration and aging involve altered plasma A β levels, epigenetic stress, tumor suppressor loss, and shortening of telomere length. These factors are also associated with vascular, metabolic, and psychological risk factors for neurodegenerative disorders (Si et al. 2021; Innes et al. 2018). Telomeres are essential for the maintenance of genomic integrity. Its length is considered a biological marker for stress and cellular aging. Interestingly, a significant shorter telomere in the peripheral cell leukocytes of treatment-resistant mesial temporal lobe epilepsy (mTLE) has also been reported (Miranda et al. 2020). More such studies can help in designing drugs targeting telomeres. This would be effective for the treatment of resistant cases of mTLE and other types of epilepsies. This might ultimately open the window of opportunities in exploring novel therapeutic avenues against epilepsy and in cognitive decline.

Studies performed to discover the options that can suppress the epileptic seizure symptoms by modulating cellular and molecular changes such as oxidative stress, neuroinflammation, apoptosis, and mTOR pathway inhibition have shown

metformin as a potential candidate in this aspect. The changes observed during the different phases of epileptic disease like neuronal loss are also reported to be modified by metformin (Yimer et al. 2019; Nandini et al. 2019).

There are no reports of dasatinib effects on epilepsy; however, quercetin has been reported recently to inhibit KA-induced epilepsy by microglia cell inactivation and the production of NF- κ B, TNF- α , and IL-1 β in mice (Wu et al. 2020). On the other hand, Xie et al. (2022) recently suggested that quercetin effectively protects against seizure-induced neuron death in vivo and in vitro and improves cognitive function impairment via the multiple pathways including SIRT1/Nrf2/SLC7A11/GPX4 pathway (Xie et al. 2022).

20.2.6 Amyotrophic Lateral Sclerosis

The hallmark effects of senescence have also been found to be associated with amyotrophic lateral sclerosis (ALS), and the role of removing senescent cells or restoring senescent cells back to normal function has been tried as a strategy for producing a disease-modifying ALS therapy (Maximova et al. 2021). In the experimental animal models and in postmortem human tissues of ALS, the glial senescence has been reported (Trias et al. 2019; Sivasubramanian et al. 2020). Senolytics affecting heat shock protein 90 (Hsp90) (17-DMAG), PI3K inhibition and DNA intercalation (Q), Akt inhibition (D), non-selective Bcl-2 family inhibitions (navitoclax), etc., have been studied for their role in removal of senescent cells and in the models of CNS disorders (Maximova et al. 2021). It has been observed that current senolytics targets such as Hsp90 and PI3K inhibition may risk degenerating motor neurons. On the contrary, inhibiting Bcl-2 family members may provide neuroprotection both directly and indirectly, via glial senolysis. Exploration of alternative strategies that are selectively upregulated in senescent glia in ALS are required to be identified that could be targeted without compromising motor neurons. If such strategies work out, the slowing down of ALS at early stage would be possible and may eventually extend lifespan of such patients.

20.2.7 Miscellaneous

Agents like navitoclax and BCL-X_L inhibitors like A1331852 and A1155463 have also been reported recently to possess senolytic activity (Zhu et al. 2015). Navitoclax modulates the anti-apoptotic pathways of BCL2 and BCL-XL, which are upregulated in senescent cells. The role of navitoclax as a senolytic agent has been suggested in delivering functional advantages in mice after brain radiation (Zhu et al. 2015, 2016, 2017). While BCL-2 inhibitors are associated with a number of adverse effects, A1331852 and A1155463 are relatively selective BCL-X_L inhibitors. They appear less likely to cause neutropenia than navitoclax, making them possibly better contenders for subsequent translation into clinical applications (Zhu et al. 2016, 2017).

Piperlongumine (PL) is another natural product that is reported to have multiple effects including anti-tumor activity. Wang and co-workers showed that it killed senescent human WI-38 fibroblasts preferentially when senescence was induced by different methods such as ionizing radiation, replicative exhaustion, or ectopic expression of the oncogene *Ras* (Wang et al. 2016). It killed senescent cells alone by inducing apoptosis. It further exhibited synergistic effects with navitoclax. The studies demonstrated PL as a novel lead for developing senolytic agents, but further studies are warranted to develop piperlongumine as an anti-aging agent.

Curcumin is known to suppress acute and chronic inflammation via multiple mechanisms including ROS scavenging, stimulating antioxidant defense actions, and reducing lipid peroxidation. It has also been reported to possess anticancer effects and to inhibit cell growth in tumor cell lines such as SGC7901, HL60, and Bel7402 in a dose-dependent manner (Cui et al. 2006). Following the treatment with curcumin in the telomerase-positive human glioblastoma and medulloblastoma cell lines, inhibition of telomerase activity has also been reported (Khaw et al. 2013).

The details of dasatinib, quercetin, fisetin, curcumin, and UBX1325 are available for trials on CNS disorders such as Alzheimer's disease, cognitive impairment, ALS, and neovascular age-related macular degeneration. For the remaining senolytics, the clinical trials are mainly ongoing/completed for osteoarthritis of knee, skin diseases, lymphomas, diabetic macular edema, etc. (Kirkland and Tchkonja 2020; Raffaele and Vinciguerra 2022).

20.3 Conclusion

A variety of complex mechanisms such as mitochondrial dysfunction, neuroinflammation, oxidative load, and various other homeostatic mechanisms that are otherwise compromised during cell senescence are able to initiate the neurodegeneration and other mechanisms leading to various disorders. Senolytics seem to re-establish progenitor dysfunction, reduce tissue inflammation, and improve age- and disease-related metabolic dysfunction. Hence, they are likely to delay, prevent, or suppress multiple age-related diseases and chronic conditions and improve health and lifespan in experimental animals. Therefore, it is predicted that they may also lead to interventions for humans that delay, prevent, or treat senescence- and age-related conditions. Though the effects and role of senolytics in peripheral chronic disorders such as diabetes, aging, osteoporosis, cardiovascular diseases, pulmonary fibrosis, and cancer are reported, only a few agents have been evaluated in CNS disorders. Some of them are reported to possess serious adverse effects such as navitoclax producing hematological toxicity, hence is not completely safe. Strategies like combination of agents with distinct mechanism of action may help in showing synergistic effects. Recent therapeutic approaches based on specific drug delivery strategies and use of novel biomarkers are also being required to be developed to increase the specificity and reduce the side effects of senolytics. Clinical trials are required to assess the safety, efficacy, tolerability, and specific targets of these drugs before being available for routine clinical use. It is also

predicted that most of the senolytic agents target particular types of senescent cells, based on their origin or mechanisms by which senescence was induced. Another future approach may also be to explore methods of senescence prevention, which would require further detailed research. There is still a long way to go, but senolytics might come up a completely new range of drugs targeting novel mechanism and may become therapeutic agents to serve treatment-resistant/refractory disorders.

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